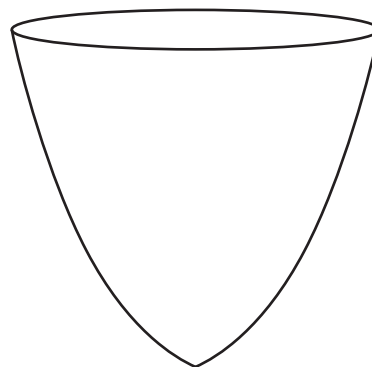
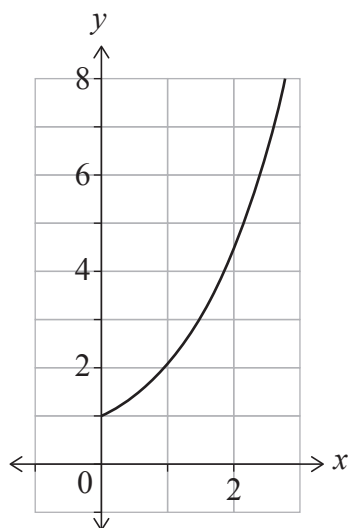


5. [Maximum mark: 5]

Consider the function $f(x) = e^{\frac{3x}{4}}$, where $x \geq 0$.

The region enclosed by the curve $y=f(x)$, the line $y=8$ and the y -axis is rotated by 2π radians about the y -axis to form a solid. This is shown in the following diagrams.



Find the volume of the solid formed.

[illegible]